*From **DisGo** to **Boogie**: Accessibility Barriers in Stanford DisGo Service*

Presenters: Ananya N., Ari B., Osose E.

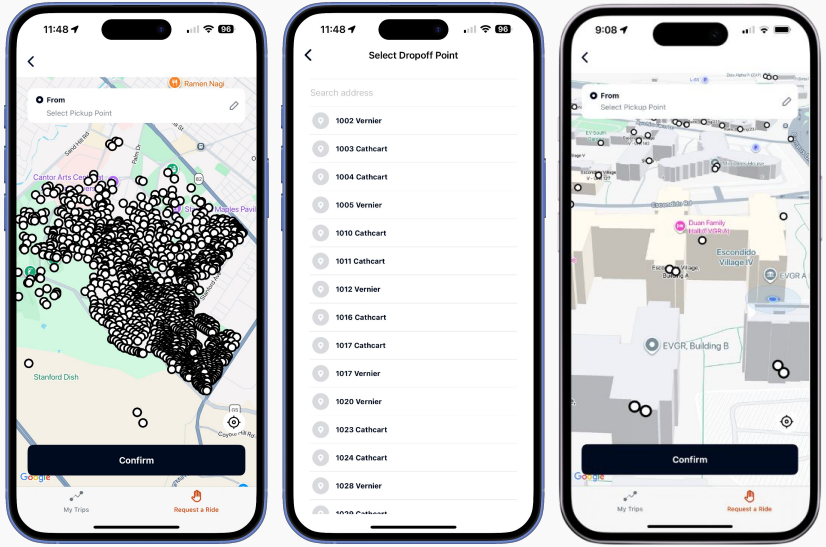
CS 377Q, Winter 2026

Contents

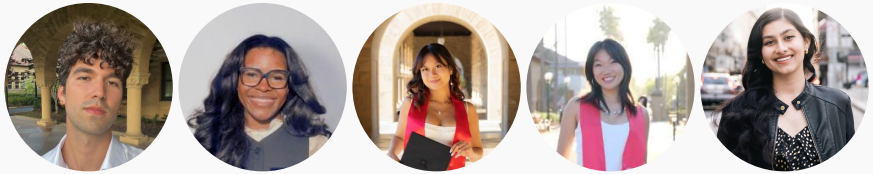
1. Intro – DisGo/Flexigo and CS 377Q
2. Interviews & Analysis
    a. Needfinding
    b. Usability (Prototype)
3. Action Items

DisGo/Flexigo

Utilized by Stanford students with mobility accommodations



CS 377Q: Design for Accessibility



- Team of 5 students (under guidance of Adjunct Lecturer John Tang)
- Examined improving usability for BLV (blind & low-vision) users
- Conducted needfinding interviews and usability tests (with our prototype)

Needfinding: Impactful Quotations from DisGo Users

Pick-Up/Drop-Off Location Selection, Locating Driver/Rider Precisely, and Opaque Ride Scheduling Failures

- "Location selection is extremely annoying… can't input a search keyword for the location… scroll through a list of 1400 items to get to your particular pickup location, and then separately, drop-off location."
- "There is a button saying my current location, when you're selecting a pickup location that hasn't quite worked for me that would… Make at least half of the process of inputting locations… Pretty much instant."
- "Most of these apps haven't invested any resources in making the maps accessible to a screen reader user like me."
- "You essentially have to work your way just by having an open, honest line of communication until either you find the cart, or the cart finds you."

Needfinding: Impactful Quotations from DisGo Users

No system for users to report service issues

What do you do when DisGo fails?

- "You don't have good workarounds in the moment, and I think the challenge is that there's not a very clear link between the disability-related orgs working for Stanford...and the people who pay DisGo...
- What I had to end up doing is raise this a number of times with Hallcon... And they were like, "yeah, sure, sorry", well, okay, and nothing really changed.
- And then I had to elevate this to someone at the DAO at Stanford, who now no longer works there. So, again, I don't even know who my current point person would be if such an issue were to emerge.
- Because this is kind of out of the ambit of the disability advisor at the OAE, and fortunately, that lady, Sheila Sanchez – amazing disability ally – she kinda got on their case, and probably had the authority to make it such that the dispatchers at DisGo would make sure that I got my rides, and would kind of manually check to make sure that the rides made it to the system, and would call a driver being like, hey, we messed up, can you quickly go and pick him up? But again, that's, like, a way that my case was handled, so that they didn't get too much fire."

Needfinding: Impactful Insights from DisGo Drivers

System Design Issues

- Not all buildings are on the search list → *confusion and friction*
- Building pins set to center of building rather than specific entrances
- App navigation assumes the cart is a car → *poor directions, inaccurate ETAs, cascading issues for subsequent scheduled rides*

Information Visibility Problems

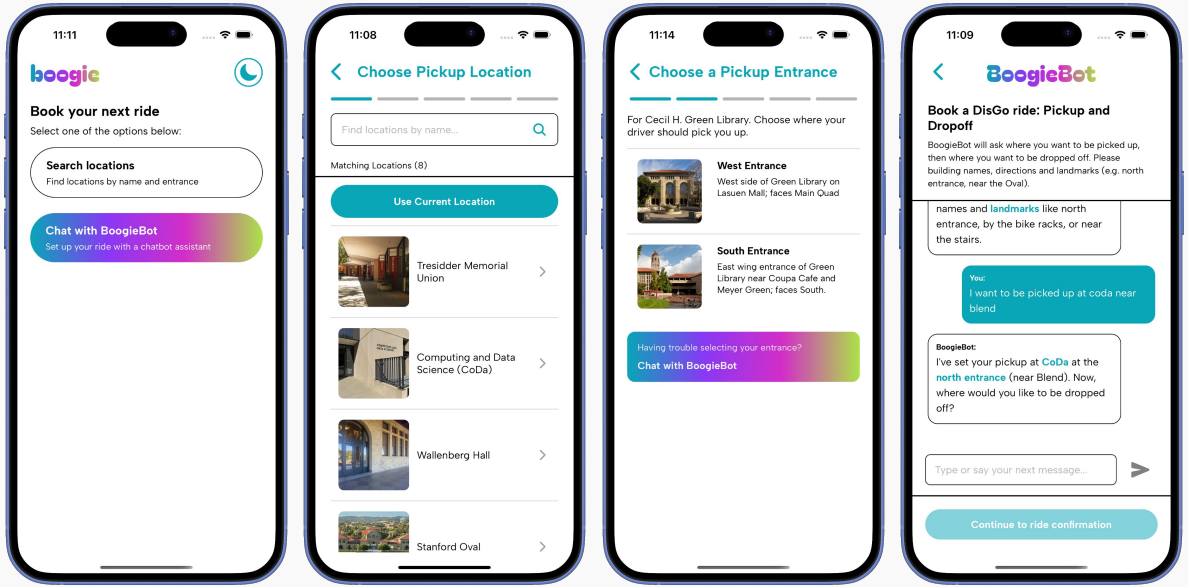
- Must click on the task for the rider's note (e.g. detailed location description) → *unintuitive, easy to miss, disruptive to driving*
- Riders can see golf cart location but drivers cannot see riders' location (one-way visibility)

Workarounds and Compensation

- Drivers memorize students' preferred pick-up/drop-off locations
- When something doesn't work, students call dispatch or message drivers via text → *extra burdens on students and drivers*
- Sometimes drivers improvise student pick-up/drop-off order

Usability (Prototype)

- Landmarks, keywords, and directions are more informative
- Specific entrances (in addition to buildings) marked for more efficient rider-cart contact
- Searchability improved
- Screen-readability improved
- Accommodation provided for light sensitivity (dark mode toggle)



Problems & Proposed Solutions

First-Tier Priorities:

*These issues **greatly negatively impact** students' ability to use DisGo successfully.*

- **Location accuracy for pick-up and drop-off**
  - Include latitude/longitude coordinates for locations → can use OpenStreetMap coordinates for ease/testing (free API)
    - This system is used by other schools contracted with Flexigo
    - Test with a set number of most frequented locations to gauge improvements
- **Searchability** ✅
- **User-grounded input**
  - Conduct regular interviews/ride-alongs with student users and drivers, offer $5-10 gift cards as compensation for feedback and time

Problems & Proposed Solutions (cont.)

Second-Tier Priorities:

*These issues **create friction** and impede the DisGo service from serving as many disabled students as possible.*

- **DisGo cart route optimization adjustment**
  - Cart currently recognized as car (staple for Flexigo) → change to bike for better routes and improved pick-up/drop-off time accuracy
- **BLV standard accessibility improvements**
  - Screen-readability can be enabled for searchable view, flexible layout for Zoom/screen magnification is doable, color contrast can be adjusted with color mode options
- **Voice-based AI conversationality**
  - Digital AI dispatcher may be helpful for rides setting less familiar destinations to provide additional information → Flexigo may have capability for this

Thank You

Appendix & Research Artifacts

- **Final Synthesis Presentation**
  - 🔗 [Google Slides]
- **Needfinding Presentation + Interview Synthesis**
  - 🔗 [Google Slides]
- **Initial Prototype Rationale**
  - 🔗 [Google Slides]
- **Usability Testing Feedback**
  - 🔗 [Google Slides]